

[BUG] Tensor does not have `\_batch\_size=None` after `map\_wires`

<https://github.com/PennyLaneAI/pennylane/issues/3660>

ROW 128

[BUG] Tensor does not have `_batch_size=None` after `map_wires` #3660

[New issue](#)[Closed](#)[#3661](#)

dwierichs opened on Jan 20, 2023

...

Expected behavior

```
>>> op = qml.PauliX(0) @ qml.PauliZ(1)
>>> print(op.batch_size)
None
>>> mapped = qml.map_wires(op, {0: 3, 1: 2})
>>> print(mapped.batch_size)
None
```



Actual behavior

```
>>> op = qml.PauliX(0) @ qml.PauliZ(1)
>>> print(op.batch_size)
None
>>> mapped = qml.map_wires(op, {0: 3, 1: 2})
>>> print(mapped.batch_size)
```



```
-----
AttributeError                                Traceback (most recent call last)
Cell In[43], line 4
      2 print(op.batch_size)
      3 mapped = qml.map_wires(op, {0: 3, 1: 2})
----> 4 print(mapped.batch_size)

File ~/repos/pennylane/pennylane/operation.py:1008, in Operator.batch_size(self)
    996 @property
    997 def batch_size(self):
    998     r"""Batch size of the operator if it is used with broadcasted parameters.
    999
   1000     The ``batch_size`` is determined based on ``ndim_params`` and the provided parameters
   (... )
   1006     int or None: Size of the parameter broadcasting dimension if present, else ``None``.
   1007     """
-> 1008     return self._batch_size

AttributeError: 'Tensor' object has no attribute '_batch_size'
```

Additional information

Recently, #3642 introduced setting the `_batch_size` of a `Tensor` object to `None` within `__init__`. However, the methods `Tensor.map_wires` (used by the PL function `map_wires`) and `Tensor.copy` do not follow this behaviour yet. Therefore the outputs of `qml.map_wires(tensor, map)` and `copy.copy(tensor)` will not have a `_batch_size` defined.

Source code

No response

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.29.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /home/david/venvs/dev/lib/python3.10/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, networkx, sc
Required-by: PennyLane-Lightning

Platform info:      Linux-5.15.0-57-generic-x86_64-with-glibc2.35
Python version:     3.10.6
Numpy version:      1.23.5
Scipy version:      1.9.3
```



Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

🔴 Set `'Tensor._batch_size'` when copying.
PennyLaneAI/pennylane

🔵 Fix `'_batch_size'` of copies and wire mappings of `'Tensor'`
PennyLaneAI/pennylane

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Participants



Zo

Author-email:
License: Apache License 2.0
Location: /home/david/venvs/dev/lib/python3.10/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, retworkx, sc
Required-by: PennyLane-Lightning

Platform info: Linux-5.15.0-57-generic-x86_64-with-glibc2.35
Python version: 3.10.6
Numpy version: 1.23.5
Scipy version: 1.9.3
Installed devices:
- default.gaussian (PennyLane-0.29.0.dev0)
- default.mixed (PennyLane-0.29.0.dev0)
- default.qubit (PennyLane-0.29.0.dev0)
- default.qubit.autograd (PennyLane-0.29.0.dev0)
- default.qubit.jax (PennyLane-0.29.0.dev0)
- default.qubit.tf (PennyLane-0.29.0.dev0)
- default.qubit.torch (PennyLane-0.29.0.dev0)
- default.qutrit (PennyLane-0.29.0.dev0)
- null.qubit (PennyLane-0.29.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.28.0)

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



dwierichs added **bug** on Jan 20, 2023



dwierichs mentioned this on Jan 20, 2023

[Fix `\_batch\_size` of copies and wire mappings of `Tensor` #3661](#)



AlbertMitjans mentioned this on Jan 20, 2023

[Set `Tensor.\_batch\_size` when copying. #3662](#)



dwierichs linked a pull request that will close this issue [Fix `\_batch\_size` of copies and wire mappings of `Tensor` #3661](#) on Jan 20, 2023



dwierichs closed this as **completed** in [#3661](#) on Jan 20, 2023

Fix `_batch_size` of copies and wire mappings of `Tensor` #3661

<> Code

Merged dwierichs merged 1 commit into `master` from `batch_size_copy_Tensor` on Jan 20, 2023

Conversation 3 Commits 1 Checks 0 Files changed 3

+18 -1



dwierichs commented on Jan 20, 2023

Contributor

Context:

The `_batch_size` attribute of `Tensor` was not copied properly when using `Tensor.map_wires` or `Tensor.__copy__`

Description of the Change:

Change that.

Benefits:

The following works:

```
>>> op = copy.copy(qml.PauliZ(0) @ qml.PauliX(1))
>>> op.batch_size
None
```



Possible Drawbacks:

N/A

Related GitHub Issues:

[#3660](#)



tests

86f524e



AlbertMitjans approved these changes on Jan 20, 2023

View reviewed changes

AlbertMitjans left a comment

Contributor

Amazing!



AlbertMitjans commented on Jan 20, 2023 • edited

Contributor

If you write: "Fixes [#3660](#)" in the PR description, the issue will be closed when this PR is merged.



1

Reviewers

AlbertMitjans



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

🔗 [BUG] Tensor does not have `_batch_size=None`

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2 participants

